Practical No - 3

Perform the data classification using classification algorithm using our R/Python.

Get the data points in form of a R vector.

→ rainfall <-

→C(799,1174.8,865.1,1334.6,635.4,918.5,685.5,998.6,784.2,985,882.8,1071)

#Convert it to a time series object.

→rainfall.timeseries <- ts (rainfall, start=c (2012,1),frequency=12)

Print the time series data.

→ print(rainfall.timeseries)

Give the chart file a name.

→ png(file = “rainfall.png”)

Plot a graph of the time series.

→ plot(rainfall.timeseries)

Save the file.

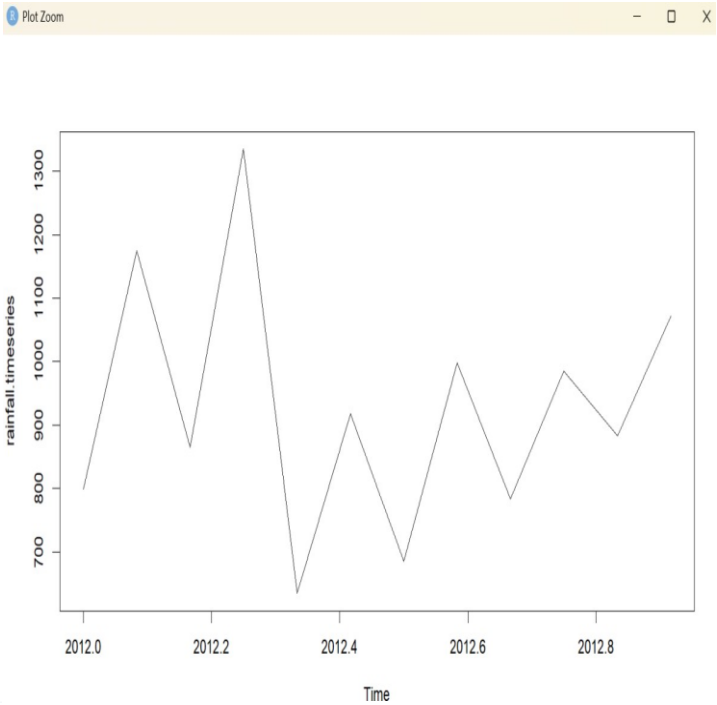
→ dev.off()

After this again plot to get chart.

→ plot(rainfall.timeseries)

OUTPUT:-

```
> # Get the data points in form of a R vector.
> rainfall <-
+   c(799,1174.8,865.1,1334.6,635.4,918.5,685.5,998.6,784.2,985,882.8,1071)
> # Convert it to a time series object.
> rainfall.timeseries <- ts(rainfall,start = c(2012,1),frequency = 12)
> # Print the timeseries data.
> print(rainfall.timeseries)
       Jan    Feb    Mar    Apr    May    Jun    Jul    Aug    Sep    Oct    Nov
2012  799.0 1174.8  865.1 1334.6  635.4  918.5  685.5  998.6  784.2  985.0  882.8
       Dec
2012 1071.0
> # Give the chart file a name.
> png(file = "rainfall.png")
> # Plot a graph of the time series.
> plot(rainfall.timeseries)
> # Save the file.
> dev.off()
null device
          1
> #After this again plot to get chart
> plot(rainfall.timeseries)
>
```



Practical No - 4

Perform the data clustering using clustering algorithm using R/Python.

k-means clustering using R.

Apply K means to iris and store result.

→newiris <- iris

→newiris$Species <- NULL

→(kc <- kmeans(newiris,3))

Compare the Species label with the clustering result.

→table(iris$Species,kc$cluster)

```
             1  2  3
setosa       0  0 50
versicolor   2 48  0
virginica   36 14  0
```

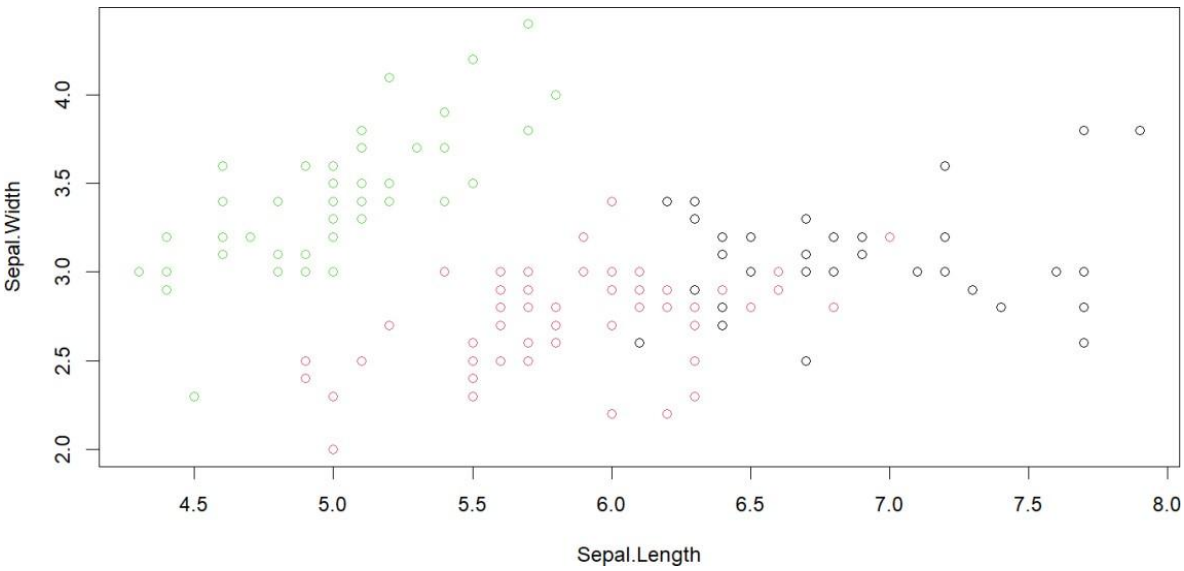
Plot the clusters and their centers.

→plot(newiris[c("Sepal.Length","Sepal.Width")],col=kc$cluster)

→points(kc$centers[,c("Sepal.Length","Sepal.Width")],col=1:3,pch=8,cex=2)

→dev.off()

Plot the clusters and their centre.

→plot(newiris[c("Sepal.Length","Sepal.Width")],col=kc$cluster)

OUTPUT:-